

Part II

The Instruments: Caps, Floors, Swaptions, and the Vol Cube

Swaps and the Forward Swap Rate

4.1 The interest rate swap

An **interest rate swap** is an exchange of a fixed rate of interest for a floating rate, on the same notional, over the same set of dates, with no exchange of principal. One party — the **fixed-rate payer** — pays a predetermined fixed rate K on each of a set of dates $T_1, \dots, T_n$ and receives, on the same dates, a floating rate that resets to the prevailing market rate for each period. The other party takes the opposite side.

Because both legs reference the same notional and the same underlying curve, a swap's value can be written entirely in terms of the discount factors and forward rates from Chapter 2, with no additional machinery.

Key result: value of a payer swap

The value at time $t \leq T_0$ of a payer swap with fixed rate K , notional $N = 1$, and floating-rate resets on accrual periods $[T_0, T_1], \dots, [T_{n-1}, T_n]$ is

$$V_{\text{payer}}(t) = \sum_{i=1}^n \tau_i P(t, T_i) (L(t; T_{i-1}, T_i) - K)$$

where τ_i is the accrual fraction and $L(t; T_{i-1}, T_i)$ the forward rate for period i , as defined in Chapter 2. This follows directly from valuing each floating cash flow as a forward-rate-driven payment and each fixed cash flow as a certain payment, discounted with the appropriate curve.

4.2 The par swap rate and the annuity

The rate K that makes the swap worth exactly zero at inception is the **par swap rate**, and it is one of the most important quantities in the entire rates market — it is what is meant, colloquially, by “the 5-year swap rate.”

Key result: the par swap rate and the annuity

Setting $V_{\text{payer}}(t) = 0$ and solving for K :

$$S(t) = \frac{\sum_{i=1}^n \tau_i P(t, T_i) L(t; T_{i-1}, T_i)}{\sum_{i=1}^n \tau_i P(t, T_i)} = \frac{P(t, T_0) - P(t, T_n)}{A(t)}$$

where

$$A(t) = \sum_{i=1}^n \tau_i P(t, T_i)$$

is the **annuity** (also called the **PVBP** — present value of a basis point, up to a scaling factor of 10^{-4} , or the **DV01** family of related quantities). The second form of $S(t)$ follows by recognising the sum of forward-rate cash flows as a telescoping series that collapses

to $P(t, T_0) - P(t, T_n)$; it is the fast way to compute a par swap rate from a discount curve without needing every individual forward rate.

The annuity plays the role, throughout everything that follows, that a discount factor plays for a single cash flow: it converts a quantity expressed in *rate units* into a quantity expressed in *money*. Every swaption price in Chapter 7 is built by multiplying an option-on-a-rate value by exactly this annuity.

Worked example 4.1 — par swap rate from a curve.

Using discount factors from a SOFR curve: $P(0, 1) = 0.9615$, $P(0, 2) = 0.9245$, $P(0, 3) = 0.8890$, with annual (simplified) payment dates and $\tau_i = 1$ for each period, find the 3-year par swap rate and the annuity.

Step 1 — the annuity.

$$A(0) = 1 \times 0.9615 + 1 \times 0.9245 + 1 \times 0.8890 = 2.7750$$

Step 2 — the par rate via the telescoping form.

$$S(0) = \frac{P(0, 0) - P(0, 3)}{A(0)} = \frac{1 - 0.8890}{2.7750} = \frac{0.1110}{2.7750} = 0.04000$$

Step 3 — express as a rate.

$$S(0) = 4.00\%$$

Sanity check. $A(0) = 2.775$ is close to $n = 3$ (the number of annual periods) as expected for discount factors not far below 1, and 4.00% sits sensibly between the 1-year and 3-year implied zero rates one would back out of these discount factors. Both magnitudes are plausible for a mid-2020s SOFR curve.

Interpretation. This 3-year par rate of 4.00% and annuity of 2.775 are the two numbers a swaption desk needs before it can price a single swaption on this underlying — Chapter 7 uses exactly this pair.

PITFALL

The annuity $A(t)$ is not a constant — it is itself a random quantity that moves as the curve moves, since it is built from discount factors to future dates. When later chapters describe the forward swap rate $S(t)$ as a “martingale under the annuity measure,” this is precisely why a dedicated measure is needed: $S(t)$ is only a clean, driftless random process when values are measured *in units of the annuity itself*, not in cash. Forgetting that the annuity is stochastic and treating it as if it were a fixed discounting constant is a common source of subtly wrong swaption Greeks.

ON THE DESK

Every swap trader’s mental shorthand for risk is a DV01 or PVBP: how much does the position gain or lose per one-basis-point parallel shift in rates. This is, up to sign and scaling convention, the annuity $A(t)$ from this chapter, multiplied by the notional and by 10^{-4} . When a swaptions desk quotes a price “per annuity” or discusses “annuity risk” separately from “rate risk,” they are decomposing the swaption’s total sensitivity into the piece coming from the forward rate moving and the piece coming from the dis-

counting/annuity itself moving — a decomposition that only makes sense once you have internalised that these are two distinct sources of randomness in the same instrument.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Write the value of a payer swap in terms of discount factors and forward rates.
2. Derive the par swap rate as the fixed rate that zeroes the swap's value, and compute it numerically from a short discount curve.
3. State what the annuity represents economically (a PVBP/DV01-style quantity) and explain why it must be treated as stochastic, not constant, in an options pricing context.

